

Oliver Dippel

Research Scientist — Reinforcement Learning & Foundation Models
St Gallen, Switzerland

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Profile

Ph.D. researcher in reinforcement learning with a focus on foundation-model-driven decision making, adaptive control, and learning from feedback. Experienced in developing principled RL algorithms, transformer-based policies, and scalable experimental systems bridging theory and practice. Strong interest in alignment, generalization, and robust agent behavior, with research contributions spanning belief-aware policies, uncertainty-aware optimization, and feedback-driven post-training.

Experience

Doctoral Researcher — Reinforcement Learning & Foundation Models Dec 2021 – Present
University of Liverpool Liverpool, United Kingdom

- Developed reinforcement learning algorithms for adaptive decision-making under uncertainty, with emphasis on policy optimization, generalization, and stability.
- Proposed transformer-based policy priors enabling in-context adaptation and belief-aware updates without parameter retraining.
- Investigated reward design, posterior regularization, and exploration–exploitation trade-offs, with direct relevance to RLHF and agent alignment.
- Designed large-scale, reproducible experimentation pipelines in PyTorch and JAX, supporting systematic ablations and long-horizon training.
- Authored peer-reviewed publications and collaborated with researchers across RL, control, and probabilistic ML.

Research Engineer — LLM Agents & Generative Systems Sep 2023 – Dec 2024
ZeroAI Liverpool, United Kingdom

- Built LLM-based agents for structured reasoning, decision support, and policy summarization in safety-critical settings.
- Integrated language-model reasoning with downstream planning and control components, emphasizing interpretability and failure analysis.
- Applied diffusion and generative models for trajectory forecasting and model-based decision support.
- Worked closely with research and engineering teams to evaluate agent behavior in closed-loop simulations.

Freelance Statistics Tutor Jan 2021 – Present
statistiknachhilfe.ch Zurich, Switzerland

- Supervised Bachelor's and Master's theses with a focus on experimental design, statistical rigor, and reproducibility.
- Taught probability, statistical inference, and applied modeling using R and Python.

Data Science Intern — Applied ML Systems Jun 2020 – Dec 2020
Novartis Pharma AG Basel, Switzerland

- Developed data pipelines and analysis tools for ML-driven decision support in regulated environments.

Research Projects

Heuristic Transformer: Belief-Augmented In-Context Reinforcement Learning

- Introduced belief-augmented transformer policies for in-context RL, enabling rapid adaptation from partial and delayed feedback.
- Analyzed stability, variance, and reward misspecification, connecting Bayesian RL with modern foundation-model training.

Feedback-Driven Policy Optimization

- Studied posterior-regularized objectives and reward shaping for learning under noisy and underspecified feedback.

Skills

Reinforcement Learning, Deep RL, Bayesian RL, Foundation Models, Transformers, RLHF Concepts, Policy Optimization, Probabilistic ML, Generative Models, Python, PyTorch, JAX, Distributed Training, Experiment Tracking

Education

Ph.D. Electrical, Electronic & Control Engineering — University of Liverpool	2021 – 2025
Thesis: Bayesian Reinforcement Learning for Control of Continuous Industrial Processes	
M.Sc. Applied Statistics — University of Göttingen	2017 – 2020
Thesis: Deep Unsupervised Data Compression on Life Insurance Client Data	
B.A. Economics — University of Göttingen	2014 – 2017
Thesis: Choice of Smoothing Parameters in Penalized Linear Models	

Publications

Heuristic Transformer: Belief-Augmented In-Context Reinforcement Learning, 2025
Contextual Transformers for Goal-Oriented Reinforcement Learning, 2024
Deep Reinforcement Learning for Continuous Control of Material Thickness, 2023

Languages

German (native), English (fluent), French (intermediate)